

Unit V

STRUCTURE OF POLYMERS

Introduction

- Polymers are **natural or synthetic** resins or their compound which can be molded, extended, cast or used as films or coatings.

Naturally occurring polymers

- Those derived from plants and animals.

Synthetic organic polymers

- Do not occur in nature are prepared artificially
- Prepared from coal and petroleum products

Classification

- Polymers are most commonly classified as (a) thermoplastics and (b) thermosetting polymer.

Thermoplastic Materials

- The polymeric materials which soften on the application of heat with or without pressure, but require cooling to set them to shape are called Thermoplastic Materials. These can be heated and cooled any number of times, but should not be heated above the decomposition temperature.
- Some important thermoplastic materials are Polythene, Polyvinyl chloride, Polystyrene etc.

Thermosetting Materials

- Polymers which require heat and pressure to mold them into shape and become permanently hard during shaping are called Thermosetting Materials. These materials cannot be re-softened once they are set and hardened.
- Some important thermoplastic materials are Phenol Formaldehyde, Epoxy Resins, and Polyesters etc.

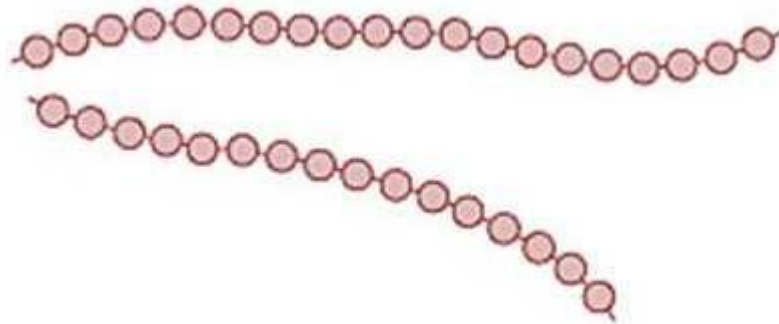
Molecular Structure

- The physical characteristics of a polymer depend not only on its molecular weight and shape but also on differences in the structure of the molecular chains.

Linear polymers

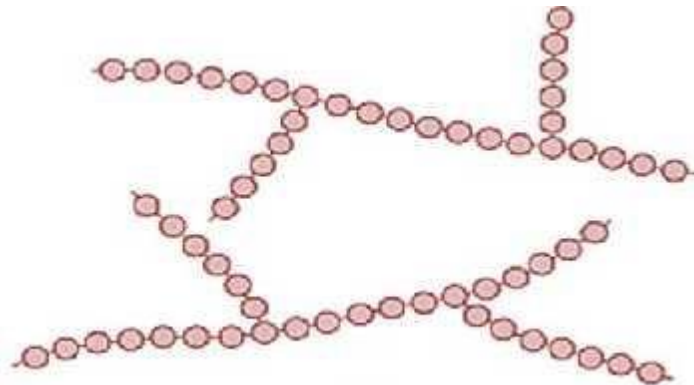
- The linear polymers are those in which repetitive molecular units are joined together end to end in single chains.

common polymers that form with linear structures are polyethylene, polystyrene, nylon, and the fluorocarbons



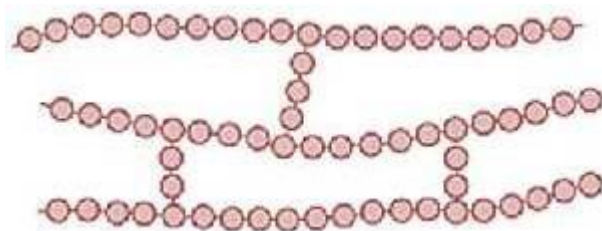
Branched Polymers

- These polymers are characterized by side-branch chains that are connected to the main ones. The chain packing efficiency is reduced with the formation of side branches, which results in a lowering of the polymer density.



Cross linked Polymers

These polymers are characterized with adjacent linear chains that are joined one to another at various positions by covalent bonds.



Network Polymers

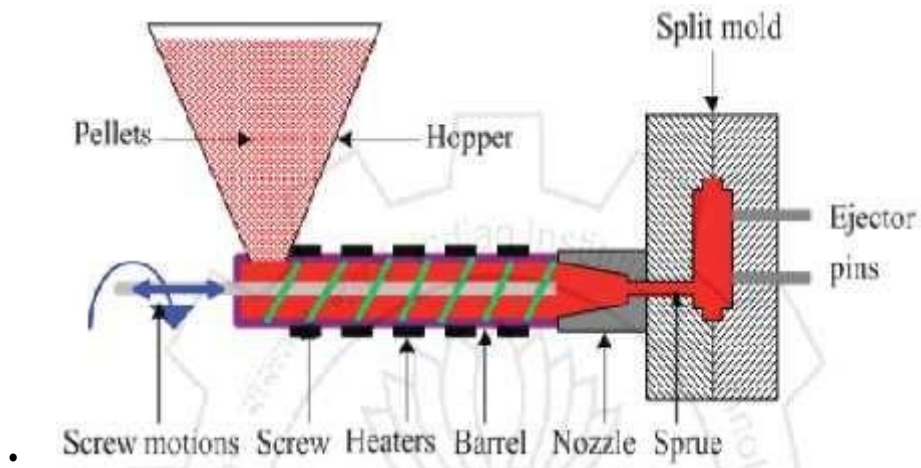
- These polymers are formed when multifunctional monomers forming three or more active covalent bonds, make three-dimensional networks. A polymer that is highly cross linked may also be classified as a network polymer.



Thermoplastic polymers	Thermosetting polymers
(1) These soften and melt on heating.	These do not soften on heating but rather become hard in case prolonged heating is done these start burning.
(2) These can be remoulded recast and reshaped.	These can not be remoulded or reshaped.
(3) These are less brittle and soluble in some organic solvents.	These are more brittle and insoluble in organic solvents.
(4) These are formed by addition polymerisation.	These are formed by condensation polymerisation.
(5) These have usually linear structures. Ex. Polyethylene, PVC, teflon.	These have three dimensional cross linked structures. Ex. Bakelite, urea, formaldehyde, resin.

Injection Molding

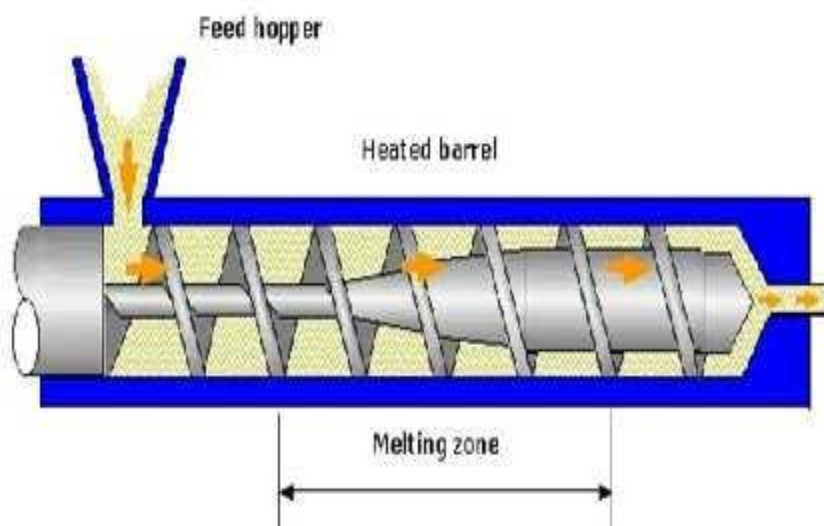
- In this process, plastic granules are heated and forced under pressure into a die cavity of desired shape.
- This process is well suited for producing true three-dimensional shapes such as bottles, toys etc, which require fine details like holes, snaps and surface details.



Extrusion

Extrusion is one of the few continuous plastic processes, which is used to produce sheet, film, long length with a profiled cross section and fiber. The chief concern with *extrusion of polymers* is the die swell and the orientation. In die swell, the extrudate swells to a size greater than the die from which it just exited.

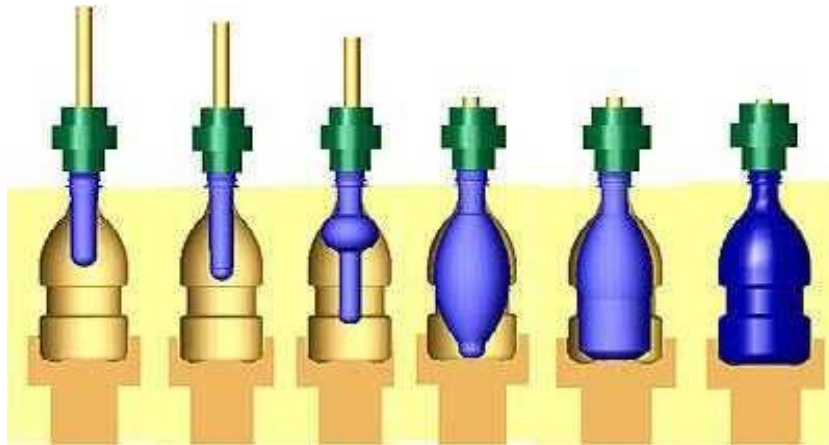
Thus the design must compensate for the swell. Polymer molecules become highly oriented in one or two directions as a result of the strongly oriented flow inherent in the extrusion process. Control of orientation can improve the property of the material typical set-up of polymer extrusion process.



Blow Molding

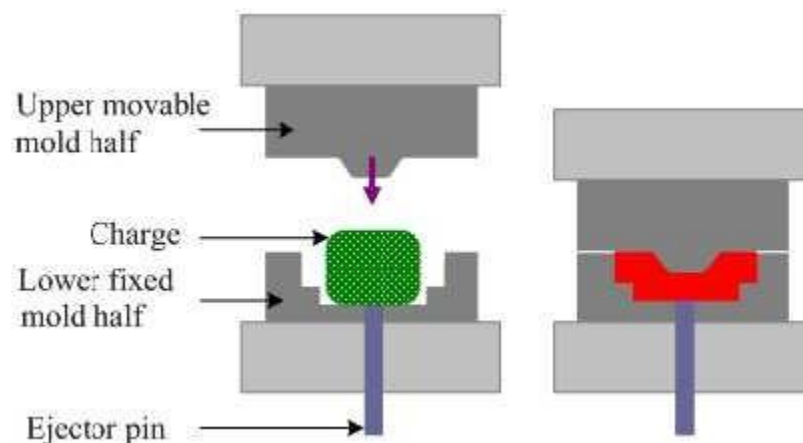
In blow molding, a hot tube of plastic material is placed between two halves of the mold. The mold is then closed and air or inert gas is blown at a pressure of 20 – 40 MPa which expands the hot tube outward to fill the mold cavity. The part cools, hardened and is ejected from mold.

The process produces a part that is dimensionally defined on its external dimension. The process does not lend itself to incorporate design details such as holes, corners, narrow ribs, etc. The blow molding process is very fast and can produce part very economically. The process can be accomplished manually or by semi-automatic or automatic machines.



Compression molding

In this technique, a preform of a typical polymer is placed in a heated mold cavity and a plunger applies pressure to force the polymer to fill the mold cavity. The material is then allowed to cure and ejected from the mold. As the amount of flow is much lesser than the same used in injection molding, the level of residual stress in the part is low. Parts made in this way would have sprues and runners which must be trimmed



ADDITIVES IN PLASTICS

- Higher polymer material is mixed with 4 to 10 ingredients each of which imparts some useful property to the moulded products.

Resins:

- Holds the different constituents together
- **Example:** Linear polymers with low molecular weights.
- Resins form 40-50% of moulding mixture

Plasticisers:

- To increase the plasticity and flexibility
- It is in the form of liquids.
- **Example:** Castor oil, Dibutyl phthalate, Alkyl & Aryl. Phosphate etc.,

Fillers:

- To impart valuable service properties such as strength, resistance, finish and workability.
- Reducing cost, shrinkage on setting and brittleness.
- **Example:** Wood flour, asbestos, china clay, talc, gypsum, saw dust.
- Fillers form 60% of plastics

Lubricants:

- To facilitate the moulding operations.
- Gives glossy finish
- **Example:** Waxes, soaps, Oleates, stearates

Catalyst/accelerators:

- To accelerate the polymerisation (added only in thermosetting resins)
- **Example:** Hydrogen peroxide, Benzoyl peroxide,
- Metals (Ag, Cu & Pb)

Stabilizers:

Rubber with carbon black when used as stabilizer gives protection against ultra violet rays